

APPLICATION SOFTWARE

With the proper software, a computer is a valuable tool. Software allows users to create letters, memos, reports, and other documents; design Web pages and diagrams; draw and alter images; record and enhance audio and video clips; prepare and file taxes; play single player or multiplayer games; compose e-mail messages and instant messages; and much more. To accomplish these and many other tasks, users work with application software. Application software consists of programs designed to make users more productive and/or assist them with personal tasks. Application software has a variety of uses:

1. To make business activities more efficient
2. To assist with graphics and multimedia projects
3. To support home, personal, and educational tasks
4. To facilitate communications

The table in Figure 3-1 categorizes popular types of application software by their general use. Although many types of communications software exist, the ones listed in Figure 3-1 are application software oriented. Successful use of

application software often requires the use of one or more of the utility programs identified at the bottom of Figure 3-1.

The categories of application software in Figure 3-1 are not mutually exclusive. Software listed in one category may be used in other categories. For example, desktop publishing programs, which are categorized as graphics and multimedia software, often are used for business or personal reasons.

Application software is available in a variety of forms: packaged, custom, Web-based, open source, shareware, freeware, and public domain.

- **Packaged software** is mass-produced, copyrighted retail software that meets the needs of a wide variety of users, not just a single user or company. Microsoft Office 2007 and Adobe Photoshop are examples of packaged software. Packaged software is available in retail stores or on the Web.
- **Custom software** performs functions specific to a business or industry. Sometimes a company cannot find packaged software that meets its unique requirements. In this case, the company may use programmers to develop tailor-made custom software, which usually costs more than packaged software.

FOUR CATEGORIES OF APPLICATION SOFTWARE

(and some popular utility programs)

APPLICATION SOFTWARE	1 Business	2 Graphics and Multimedia	3 Home/Personal/Educational
	• Word Processing • Spreadsheet • Database • Presentation Graphics • Note Taking • Personal Information Manager (PIM) • Personal Mobile Device Business Software • Software Suite • Project Management • Accounting • Document Management • Enterprise Computing	• Computer-Aided Design (CAD) • Desktop Publishing (for the Professional) • Paint/Image Editing (for the Professional) • Photo Editing (for the Professional) • Video and Audio Editing (for the Professional) • Multimedia Authoring • Web Page Authoring	• Software Suite (for Personal Use) • Personal Finance • Legal • Tax Preparation • Desktop Publishing (for Personal Use) • Paint/Image Editing (for Personal Use) • Photo Editing and Photo Management (for Personal Use) • Clip Art/Image Gallery • Video and Audio Editing (for Personal Use) • Home Design/Landscaping • Travel and Mapping • Reference and Educational • Entertainment
	4 Communications		
	• Web Browser • RSS Aggregator	• E-Mail • Blogging	• Instant Messaging • Newsgroup/Message Board • Chat Room • FTP • Text Messaging • VoIP • Video Conferencing
	Popular Utility Programs		
	• Antivirus • File Compression	• Personal Firewall • Backup • Spyware Remover • Media Player	• Internet Filters • CD/DVD Burning • File Manager • Personal Computer Maintenance

FIGURE 3-1 The four major categories of popular application software are outlined in this table. Communications software often is bundled with other application or system software. Also identified at the bottom of the figure are widely used utility programs.

- **Web-based software** refers to programs hosted by a Web site. Users access and interact with Web-based software from any computer or device that is connected to the Internet. Many Web sites provide free access to their programs; some charge a fee. Types of Web-based software include e-mail, word processing, tax preparation, and game programs. Web-based programs are discussed in more depth later in the chapter.
- **Open source software** is software provided for use, modification, and redistribution. This software has no restrictions from the copyright holder regarding modification of the software's internal instructions and its redistribution. Open source software usually can be downloaded from the Internet, often at no cost.
- **Shareware** is copyrighted software that is distributed at no cost for a trial period. To use a shareware program beyond that period, you send payment to the program developer. Shareware developers trust users to send payment if software use extends beyond the stated trial period. In some cases, a scaled-down version of the software is distributed free, and payment entitles the user to the fully functional product.
- **Freeware** is copyrighted software provided at no cost by an individual or a company that retains all rights to the software. Thus, programmers typically cannot incorporate freeware in applications they intend to sell. The word, free, in freeware indicates the software has no charge.
- **Public-domain software** has been donated for public use and has no copyright restrictions. Anyone can copy or distribute public-domain software to others at no cost.

Thousands of shareware, freeware, and public-domain programs are available on the Internet for users to download. Examples include communications programs, graphics programs, and games. These programs usually have fewer capabilities than retail programs.

After you purchase or download software, you install it. During installation, the program may ask you to register and/or activate the software. Registering the software is optional and usually involves submitting your name and other personal information to the software manufacturer or developer. Registering the software often entitles you to product support. *Product activation* is a technique that some software manufacturers use to ensure the software is not installed on more computers

than legally licensed. Usually, the software does not function or has limited functionality until you activate it via the Internet or telephone. Thus, activation is a required process for programs requesting it. Registering and/or activating the software also usually entitles you to free program updates for a specified time period, such as a year.

The Role of System Software

System software serves as the interface between the user, the application software, and the computer's hardware (Figure 3-2). To use application software, such as a word processing program, your computer must be running system software — specifically, an operating system. Four popular personal computer operating systems are Windows Vista, Windows XP, Linux, and Mac OS X.

Each time you start a computer, the operating system is *loaded* (copied) from the computer's hard disk into memory. Once the operating system is loaded, it coordinates all the activities of the computer. This includes starting application software and transferring data among input and output devices and memory. While the computer is running, the operating system remains in memory.

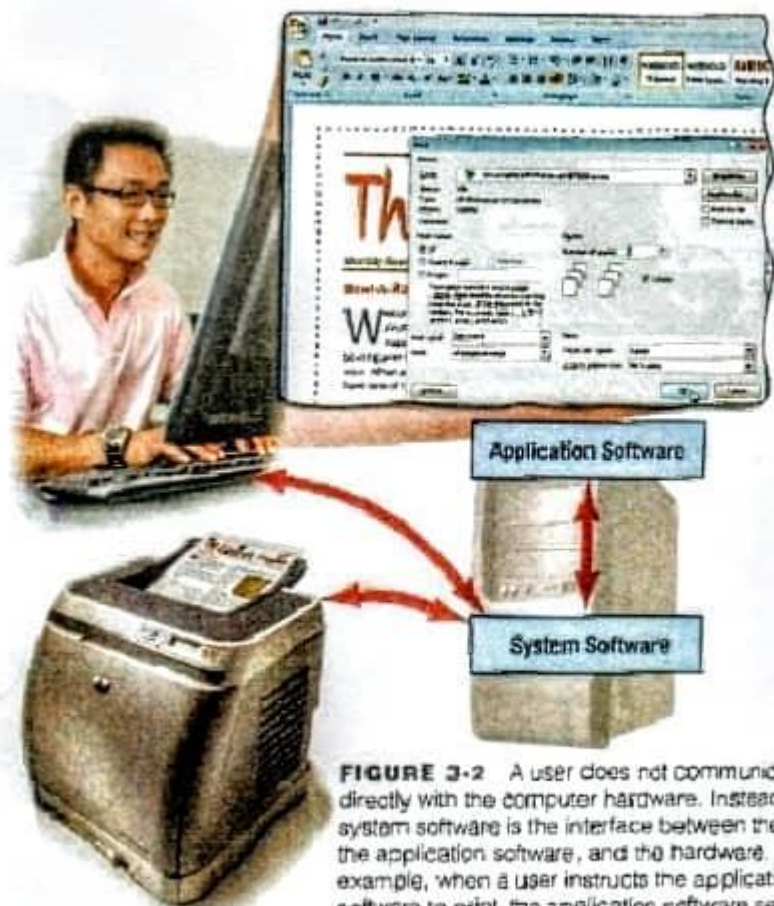


FIGURE 3-2 A user does not communicate directly with the computer hardware. Instead, system software is the interface between the user, the application software, and the hardware. For example, when a user instructs the application software to print, the application software sends the print instruction to the system software, which in turn sends the print instruction to the hardware.